

Review Article

A Systematic Literature Review of Information Security Practices in Higher Education Contexts

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Information security in institutions of higher learning is a persistent challenge that requires immediate attention. The plethora of reported information security breaches in higher education institutions (HEIs) corroborates this. This study reviews 358 articles from reputable journals and databases, synthesized following the preferred reporting items for systematic reviews and meta-analyses framework, to examine the information security practices of HEIs in the last 5 years. This study reveals that information security in HEIs is best advanced through a holistic strategy that integrates adaptive policy frameworks, continuous and context-specific training, resilient organizational culture, and judicious deployment of technological innovations. While existing approaches have improved compliance and risk mitigation, persistent gaps in implementation, resource allocation, and cultural alignment limit their overall impact. By synthesizing insights from policy, training, cultural, and technological perspectives, this research underscores the need for socio-technical approaches that recognize human behavior as a pivotal determinant of institutional resilience. Furthermore, the analysis highlights how emerging technologies, such as AI-driven threat detection and blockchain, can only achieve their transformative potential when embedded within inclusive governance structures and culturally supportive environments. The contribution of this study lies in demonstrating that sustainable cybersecurity in higher education requires a shift from fragmented interventions toward integrated, system-wide frameworks that harmonize people, processes, and technology. Such an approach not only strengthens protection of academic and research assets but also provides a foundation for long-term adaptability in the face of evolving cyber threats. The study advances an information security conceptual model and identifies gaps for areas of further research.

Keywords: information security; information security behavior; information security measures; threats; vulnerabilities

1. Introduction

Research on information security practices in higher education institutions (HEIs) has emerged as a critical area of inquiry due to the increasing reliance on digital technologies and the consequent exposure to sophisticated cyberattacks. The evolution from traditional to digital and hybrid learning environments, accelerated by the COVID-19 pandemic, has expanded the attack surface for cybercriminals targeting

sensitive academic, financial, intellectual property, research, and personal data for both faculty and students at HEIs [1–4].

HEIs face unique challenges stemming from open networks, bring your own device (BYOD) policies, diverse user populations, research collaborations, and complex information systems, making them prime targets for information security threats and attacks [5–7]. Also, the constant influx of new students, faculty, and staff, each with varying levels of

technical expertise and security awareness, further compounds the challenge of maintaining a strong security posture [8].

Moreover, the decentralized IT infrastructure common in many universities, where individual departments or research groups manage their own servers and websites, further complicates security efforts and expands the attack surface [9–11], and yet the cybersecurity landscape is constantly evolving, with new threats emerging regularly, placing a significant burden on organizations to stay ahead, which is not the case with HEIs [12]. A separate survey examining information security within HEIs indicated that 91% of these establishments reported security breaches in the preceding year, a proportion notably higher than that observed in the business sector [13].

Consequently, the practical significance of securing these environments is underscored by the rising frequency of data breaches and ransomware campaigns, which cause huge financial losses, threaten institutional operations, and damage stakeholder trust [14, 15]. For example, studies have shown that data breaches can impose considerable financial strains, averaging \$3.92 million and impacting around 25,575 records annually [16]. In fact, according to a BlueVoyant study encompassing 2702 universities across 43 countries, ransomware attacks targeting universities doubled between 2019 and 2020, thus becoming the foremost cyber threat, with the average cost of such attacks reaching \$447,000 (BlueVoyant [17]).

Existing research often focuses on technological solutions but lacks comprehensive system-wide perspectives that integrate organizational culture, human factors, and policy compliance [18–20]. Moreover, there is a notable gap in understanding how cybersecurity awareness and training influence user behavior and policy adherence within these institutions [21–23]. Controversies persist regarding the effectiveness of current training programs and the extent to which institutional commitment impacts security outcomes [24]. Failure to address these gaps risks continued vulnerabilities, data breaches, and reputational damage [3, 22].

Furthermore, while some studies emphasize technological defenses, others highlight the critical role of human behavior and organizational culture in cybersecurity resilience in HEIs [19, 25, 26]. The effectiveness of technical controls versus human-centric strategies remains a subject of debate, with evidence suggesting that neglecting either approach can lead to exacerbated vulnerabilities [27, 28]. This oversight results in heightened susceptibility to attacks, compromised data integrity, and operational disruptions [3, 29].

This systematic review, therefore, aims to synthesize extant knowledge that underscores that information security within HEIs is a multifaceted and evolving concern, deeply shaped by policy frameworks, training effectiveness, historical trajectories, organizational culture, and technological innovations. By addressing these identified gaps, this review endeavors to contribute to the development of comprehensive, context-specific information security frameworks, thereby enhancing the protection and resilience of HEIs [30–32]. This study aims to address the following research inquiries pertaining to information security practices within HEIs:

1. To what extent do HEIs develop and implement comprehensive and adaptable cybersecurity policies?
2. How effective are cybersecurity training programs in improving compliance and behavior among higher education stakeholders?
3. How have information security practices in HEIs progressed across different historical phases?
4. What role does organizational culture play in shaping cybersecurity behavior and policy compliance within HEIs?
5. How are emerging technological innovations transforming cybersecurity practices in HEIs?

This paper is organized as follows: Section 2 provides a comprehensive overview of existing systematic literature reviews concerning information security in HEIs; Section 3 delineates the methodology employed; Section 4 synthesizes the included and reviewed literature; Section 5 discusses the research questions, Section 6 undertakes the thematic review and comparative analysis, Section 7 discusses the theoretical and practical implication, Section 8 discusses the limitations of the study, and Section 9 discusses the conclusion, gaps, and future recommendations.

2. Previous Systematic Review Studies

In their study, Hina and Dominic [33] examined management, awareness, and culture in relation to information security policy compliance. According to their research, the adherence of HEIs to these policies was not very high. They suggest that thorough, easily comprehensible information security policies, focused awareness campaigns, and consistent monitoring procedures are necessary for strong institutional governance.

While this study proposes that enhanced compliance in HEIs can be achieved through clearer information security policies, awareness campaigns, and monitoring mechanisms, these strategies may not fully address the underlying structural and cultural complexities characteristic of these environments. Low adherence to security policies often stems from systemic issues, such as limited resources, decentralized governance, and the academic values of autonomy and openness, rather than a mere lack of clarity or awareness [34–36]. Furthermore, an overemphasis on monitoring could erode trust and foster resistance, counteracting the intended security culture [37, 38]. Consequently, a more effective and sustainable approach involves not only refining policies and campaigns but also integrating security practices into academic workflows, adopting user-friendly technologies, and cultivating a collective sense of responsibility that aligns security with the core academic mission of higher education [39, 40].

Similarly, Imbaquingo-Esparza et al. [41] underscore the necessity for HEIs to adopt a stringent information security paradigm, which entails the establishment of exhaustive security policies, the deployment of resilient technical safeguards, and the execution of continuous, critical security assessments. Nevertheless, in spite of the significance of these measures, consistent policy adherence and comprehensive security are

frequently difficult to achieve due to the openness and dynamic technological environment of HEIs, as well as the growing sophistication of cyber threats.

Another study by Merchan-Lima et al. [42] focused on effective information security management practices for protecting HEIs' information assets from intrusions and attacks. They propose that HEIs implement information security management through the application of framework functions, implementation phases, infrastructure services, and security measures to their assets. Additionally, they suggest that institutions should employ strategies such as analysis, evaluation, and validated procedures that consider their operational architecture. However, the comprehensive implementation of such practices often faces significant challenges, including resource constraints, a shortage of skilled personnel, and the complexity of integrating new security frameworks within diverse existing IT infrastructures.

While these studies identified several security challenges influencing the implementation of security tools and measures in HEIs, they did not comprehensively examine the specific technical and human threats, critical security priorities, or the interplay between technical and human threats and measures prevalent in real-world security events characteristic of these environments.

3. Systematic Literature Review Methodology

To synthesize and expand the existing knowledge base, this study's research design entails a two-pronged approach to build upon the existing knowledge base. First, a systematic literature search is conducted to uncover relevant scholarly sources, as the rigor of a literature review hinges on the quality of the search process [43, 44]. Second, the retrieved articles are analyzed using predefined criteria to extract the key themes, knowledge gaps, and future research directions.

3.1. Literature Search Process

3.1.1. Databases and Sources. The literature search was conducted across multiple bibliographic databases and gray literature sources to maximize coverage and minimize publication bias. Selected sources included the following:

- Web of Science (core collection);
- Scopus;
- IEEE Xplore;
- ACM Digital Library;
- JSTOR;
- SpringerLink;
- Wiley Online Library;
- Emerald Insight;
- Taylor & Francis Online;
- Sage Journals;
- ScienceDirect (Elsevier);
- Google Scholar (for supplementary and forward citation searching);
- ProQuest Dissertations & Theses;

- Institutional repositories and conference proceedings for major cybersecurity and higher education conferences;
- Relevant organizational and policy reports (e.g., EDUCAUSE, national data protection authorities).

3.1.2. Search Terms and Strategy. A structured search string was tailored to each database, using Boolean operators, truncation, and proximity operators. The following master search string (adapted to each database's syntax) was used as a starting point:

("information security" OR cybersecurity OR "data security" OR "privacy") AND ("higher education" OR university OR college OR "tertiary education" OR "post-secondary") AND (practice * OR policy * OR governance OR awareness OR training OR "risk assessment" OR technol * OR control * OR "incident response" OR "access control").

The search was limited to records published in the past decade, from 2015 to 2025, in English. Where feasible, filters were applied for document types (articles, conference proceedings, reviews, and technical reports). All database-specific queries were saved and documented (date searched, database, exact query string) in a search-log spreadsheet for reproducibility.

3.1.3. Deduplication and Record Management. Search results from all sources were exported in RIS or BibTeX format and imported into the Mendeley reference manager. Deduplication was performed first automatically by the reference manager and then manually by the authors to remove residual duplicates. A master record of unique citations was maintained as the starting point for screening.

3.2. Filtering Criteria as Per the PRISMA. This review adhered to the PRISMA 2020 reporting guidelines for systematic reviews. Screening comprises three sequential stages: Title/abstract screening, full-text screening, and data extraction/quality appraisal.

3.2.1. Inclusion Criteria. For this literature review, studies were included only if they satisfied all of the following:

- Focus on information security, cybersecurity, or data privacy practices as they apply to HEIs (including students, faculty, administrative systems, and research systems).
- Empirical studies (qualitative, quantitative, and mixed-methods), systematic or scoping reviews, or rigorous conceptual frameworks/policy analyses.
- Published between 2015 and 2025.
- Published in English.
- Provide sufficient methodological detail to allow appraisal (for empirical work).

3.2.2. Exclusion Criteria. Studies were excluded if they met any of the following:

- Focus exclusively on K-12, primary, or secondary education without explicit relevance to higher education.

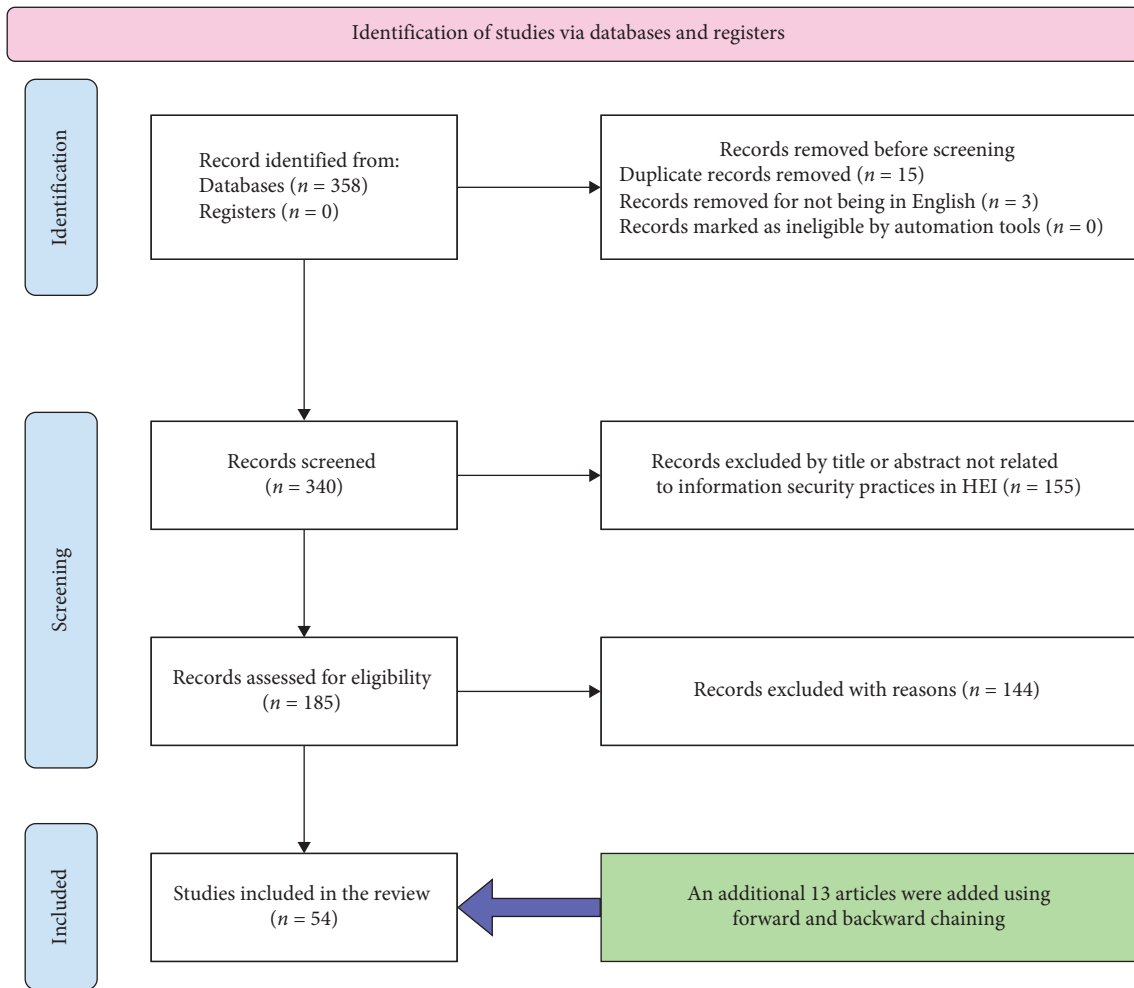


FIGURE 1: Selection of articles using PRISMA. Sources: author's own creation.

- Purely technical development articles with no evaluation or discussion of deployment/use in HEIs (unless they include a clear discussion of HEI-specific considerations).
- Editorials, opinion pieces, or short news items without empirical evidence or conceptual argumentation.
- Publications outside the date range or not available in full text.

3.2.3. Screening Procedure and Inter-Rater Reliability. The screening process involved two independent reviewers examining titles and abstracts via Silvi.ai. Initially, 358 articles were identified; however, 15 were subsequently removed due to duplication, and three articles were removed for not being written in English, resulting in 340 articles for screening. Of these, 155 articles were rejected by the reviewers because their titles and abstracts were either irrelevant to information security practices in HEIs or met the predefined exclusion criteria. Articles marked for inclusion by at least one reviewer were then advanced to a full-text screening phase, also conducted independently by two reviewers across 185 articles. Any disagreements were resolved through discussion or by a third

senior reviewer. The details of the screening process are demonstrated in Figure 1.

Inter-rater agreement was documented for both the title/abstract and full-text screening stages. This phase resulted in the exclusion of an additional 144 articles, bringing the total number of rejected articles to 317. Ultimately, 41 articles met both the inclusion and exclusion criteria and were, therefore, incorporated into the study. This rigorous screening process ensured that only the most pertinent and high-quality research informed the subsequent analysis, thereby enhancing the validity and reliability of the review's findings.

3.2.4. Backward and Forward Chaining (Citation Chasing). Beyond database searches, the literature corpus was further enriched through backward and forward citation chaining to ensure thoroughness [45, 46]. Backward citation chaining involved scrutinizing the reference lists of initial articles to identify earlier foundational studies that have shaped the current discourse. References found through this method were subsequently screened and incorporated if deemed eligible, following the established screening and extraction protocol. This technique proves especially beneficial for uncovering

seminal works potentially missed by database searches due to variations in indexing or keyword application [45, 47].

Conversely, forward citation chaining involved tracing subsequent publications that cited the initial articles, utilizing tools such as Google Scholar and Scopus. This strategy facilitated the identification of more recent contributions, guaranteeing that the review encompassed contemporary advancements and ongoing discussions concerning information security practices in higher education [48].

By integrating both backward and forward chaining methods, this study mitigated the inherent limitations of relying solely on database searches, thereby enhancing the robustness of the final dataset [45, 47, 48]. Thirteen additional articles were identified via citation chaining, culminating in a synthesis of 54 articles. These articles primarily addressed information security practices, data protection, cybersecurity policies, user awareness training, recent innovations in information security, and technological integrations within HEIs. This exhaustive approach enabled a comprehensive grasp of the complex challenges and evolving solutions within the realm of information security in academic settings.

4. Details of Articles Included in the Review

This section describes the existing research landscape concerning information security practices within HEIs, encompassing data protection, cybersecurity policies, user awareness training, historical development, recent innovations, and emerging trends. The scope of this review covers a range of research methodologies, including systematic reviews, case studies, and proposed frameworks. It addresses key aspects such as policy formulation, user conduct, technological advancements, and organizational culture across diverse global regions, including the Middle East, Africa, Asia, Europe, and the US. The predominant methodologies employed in the literature consist of qualitative analyses, surveys, comprehensive literature reviews, and mixed methods, underscoring the complex cybersecurity challenges confronting HEIs. This comparative synthesis is pivotal for addressing critical research inquiries about the efficacy of policies, the impact of training interventions, cultural determinants, and the integration of technology within academic cybersecurity contexts. The details of articles included in this study are described in the Table 1.

Figure 2, derived from Leximancer analysis, visually represents the primary research areas concerning information security practices within HEIs. Both the visual evidence from Figure 2 and existing review findings underscore the complex nature of information security in HEIs, demanding coordinated integration of data, risk, and security considerations. This complexity is compounded by the diverse technological environments prevalent in academic settings, which range from antiquated systems to advanced research infrastructure, all requiring robust and adaptable security measures [1]. Such an intricate environment, coupled with the significant volume of sensitive data, renders HEIs particularly susceptible to various cyber threats, spanning from opportunistic financial exploitation to sophisticated state-sponsored espionage [3].

Consequently, this situation is further aggravated by the transient populations within universities, as the annual rotation of students and staff introduces new security challenges and potential vectors for exploitation [3]. Similarly, the inherent openness of academic networks, designed to foster collaboration and knowledge exchange, further amplifies this susceptibility by inadvertently creating numerous entry points for malicious actors [69].

This continuous turnover necessitates ongoing information security awareness training and policy reinforcement, which often proves inadequate in practice, thereby leaving institutions vulnerable to diverse forms of cyber exploitation ([85], [86]). Therefore, this study examines policy frameworks, training effectiveness, historical evolution, organizational culture influence, and technological innovations in HEI in relation to information security.

5. Research Questions

5.1. RQ.1 To What Extent do HEIs Develop and Implement Comprehensive and Adaptable Cybersecurity Policies? Forty studies revealed that comprehensive and adaptable cybersecurity policies are crucial for HEIs; however, they are frequently underdeveloped or inadequately implemented [12, 20, 53]. These studies advocate for clearer, more flexible, and context-specific policy frameworks. Additionally, several studies underscore the importance of integrating governance, risk management, and compliance standards, such as ISO 27000, COBIT, and NIST, to improve policy efficacy [32, 74, 87]. Nevertheless, policy formulation is often impeded by organizational culture, resource limitations, and insufficient stakeholder involvement, highlighting the necessity of inclusive and transparent policy processes [24, 73].

Other frameworks suggest system-wide and human-centric approaches to policy design, emphasizing collaboration and ongoing improvement [50, 59]. Emerging challenges, including shadow IT, cloud adoption, and artificial intelligence, necessitate that information security policies remain adaptive and responsive to novel technological risks. The importance of these policies is further underscored by the rising rate of data breaches and intellectual property theft observed in colleges and universities [88].

5.2. RQ.2 How Effective Are Cybersecurity Training Programs in Improving Compliance and Behavior Among Higher Education Stakeholders? A synthesis of existing literature, encompassing 35 studies, reveals that user awareness training has a significant impact on improving cybersecurity compliance and behavior within HEIs. This enhancement is most notable when training programs are tailored and interactive, incorporating methods like gamification or simulated phishing exercises [25, 64, 69]. Furthermore, the effectiveness of such training is modulated by demographic factors, organizational culture, and the direct relevance of the content to users' specific roles and experiences [21, 23, 52]. Despite these efforts, a persistent gap often exists between theoretical knowledge and practical security behaviors, emphasizing the critical need for continuous, adaptable, and contextually relevant training initiatives [1, 66].

TABLE 1: Details of articles included in the study.

Study	Policy frameworks	Training effectiveness	Historical evolution	Organizational culture influence	Technological innovations
[20]	Identifies gaps in IS policies, calls for a culture focus	Highlights the need for human factors in training	Reviews decade-long IS trends in universities	Emphasizes organizational culture and behavior gaps	Notes emerging IS landscape, limited tech focus
[12]	Proposes system-wide institutional cybersecurity strategies	Advocates for training and awareness campaigns for culture	Chronicles cybersecurity evolution and future trends	Discusses governance and culture strengthening	Includes AI-based threat response and encryption use
[49]	Discusses best practices for educational info security	Stresses the importance of training and awareness programs	Covers 21 st -century info security development	Links security to a safe learning environment culture	Highlights advanced cybersecurity protocols and collaboration
[4]	Notes policy gaps, especially in underfunded institutions	Recommends regular cybersecurity training for stakeholders	Addresses recent digital learning security challenges	Points to privacy concerns and policy implementation gaps	Suggests encryption and multifactor authentication adoption
[50]	Presents EduCERT framework integrating governance and policies	Focuses on awareness and knowledge sharing campaigns	Framework reflects evolving cybersecurity needs	Highlights collaboration and organizational learning	Incorporates incident management and continuous improvement
[18]	Finds lack of comprehensive security policies in HEIs	Identifies awareness and culture as key compliance factors	Reviews policy compliance trends over a decade	Examines behavioral intention and organizational management	Limited focus on technological innovations
[1]	Reviews cybersecurity definitions contextualized to HEIs	Emphasizes education on password and phishing best practices	Analyzes the digital transformation's impact on security	Notes human error and insider threats in culture	Discusses hybrid learning and evolving cyber threats
[21]	Assesses policy impact on staff and student awareness levels	Measures cybersecurity knowledge and threat awareness gaps	Focuses on the recent awareness program effectiveness	Finds demographic factors influencing awareness	Limited discussion on new technologies
[25]	Case study on phishing simulation to inform policy	Demonstrates training impact via simulated attacks	Uses recent data to assess user readiness	Reveals varying security literacy among users	Employs social engineering scenarios for training
[51]	Evaluates existing policies' role in academic employee behavior	Identifies training needs to improve cyber hygiene	Reflects on current awareness levels and gaps	Highlights risk perception among academic staff	Suggests technology-based solutions for awareness
[23]	Discusses policy gaps affecting student cybersecurity resilience	Uses behavioral theory to assess awareness and attitudes	Identifies an inflection point in cybersecurity education	Explores cultural differences between undergraduate and postgraduate students	Limited focus on technological tools
[52]	Develops an awareness scale to inform policy and training	Validates behavioral factors influencing student security	Tracks the evolution of student cybersecurity behaviors	Examines demographic and cultural behavior differences	Suggests tailored cybersecurity education approaches
[53]	Proposes a comprehensive cybersecurity framework for HEIs	Integrates awareness and training in framework components	Synthesizes historical and current cybersecurity challenges	Emphasizes governance and collaboration culture	Includes incident response and technical controls
[54]	Analyzes user behavior patterns to inform policy design	Uses digital footprint analysis to enhance awareness	Applies sequential analysis to detect evolving threats	Highlights behavioral anomalies affecting security	Suggests advanced analytics for threat detection
[55]	Introduces the security culture readiness model for universities	Focuses on readiness assessment and awareness improvement	Reflects on cultural evolution in security practices	Provides actionable strategies to enhance culture	Limited direct technological innovation focus
[56]	Discusses policy implications for library security in HEIs	Highlights awareness of emerging tech in library security	Reviews recent cybersecurity threats in academic libraries	Addresses cultural challenges in technology adoption	Explores AI, blockchain, biometrics, IoT applications
[57]	Examines data security policies in financial aid operations	Links training and awareness to data security behavior	Focuses on recent institutional commitment trends	Identifies behavioral gaps in data protection culture	Limited technological innovation discussion
[58]	Reviews cloud security policies and risk management	Emphasizes training for cloud data protection	Chronicles cloud adoption and associated risks	Notes governance and transparency in culture	Highlights blockchain and AI for cloud security
[59]	Proposes a human-centric cybersecurity framework	Stresses education and awareness to reduce vulnerabilities	Reflects on recent human-factor-related cyberattacks	Focuses on poor knowledge and skills in culture	Limited technological innovation focus
[60]	Details the ISP development process and policy design	Notes the importance of policy clarity for compliance	Reviews policy evolution in HEIs	Discusses barriers related to organizational culture	Limited focus on technology integration
[32]	Reviews ICT security tools and standards in HEIs	Notes training as part of security tool adoption	Summarizes decade-long ICT security trends	Highlights organizational and physical security culture	Covers ISO 27000, CIS controls, and technical tools

TABLE 1: Continued.

Study	Policy frameworks	Training effectiveness	Historical evolution	Organizational culture influence	Technological innovations
[61]	Proposes a conceptual security awareness framework	Emphasizes user awareness as a protective technique	Reflects ongoing development of awareness programs	Identifies key dimensions affecting user behavior	Limited technological innovation discussion
[62]	Examines ICT policy and security mechanisms	Highlights policy gaps and human factor vulnerabilities	Discusses pre- and post-remote learning security changes	Emphasizes end-user errors and culture	Suggests increased training and improved practices
[63]	Describes human-centric cybersecurity program implementation	Focuses on training and management infrastructure	Reflects on program development and deployment	Addresses organizational culture in security	Limited technological innovation focus
[64]	Evaluates malware awareness program effectiveness	Demonstrates training impact on student knowledge	Focuses on recent educational interventions	Promotes a culture of cyber-resilience	Suggests ongoing development of educational initiatives
[3]	Synthesizes cybersecurity risk literature in HEIs	Notes the scarcity of empirical risk management studies	Reviews 12 years of cybersecurity risk evolution	Highlights organizational vulnerabilities	Limited focus on technological innovations
[41]	Reviews information security issues and policies	Links organizational culture to security challenges	Covers recent security issues in HEIs	Discusses the policy and culture relationship	Limited technological innovation discussion
[65]	Investigates shadow IT and related policy challenges	Notes need for awareness on unauthorized IT use	Highlights recent shadow IT trends in HEIs	Discusses cultural acceptance and risks	Addresses risks from outdated software and visibility gaps
[66]	Assesses mobile security awareness among students	Links knowledge to behavioral intentions	Reflects on recent mobile technology adoption	Notes the gap between knowledge and behavior	Limited technological innovation focus
[67]	Investigates the benefit realization of IS policies	Uses quantitative methods to assess policy impact	Tracks policy adoption and benefits over time	Highlights organizational commitment and culture	Limited technological innovation discussion
[68]	Surveys security awareness and training in Middle East HEIs	Identifies regional training program levels	Reviews historical awareness program development	Notes cultural factors in awareness	Limited technological innovation focus
[69]	Analyzes gamification in cybersecurity education	Evaluates training effectiveness via gamified methods	Discusses recent educational innovations	Highlights engagement culture in training	Focuses on gamification technology integration
[70]	Presents a model for managing organizational cyber behavior	Emphasizes internalized responsibility in culture	Reflects on behavioral management evolution	Highlights user awareness and IT capabilities	Limited technological innovation focus
[19]	Reviews human and contextual factors in cybersecurity	Synthesizes behavioral and socio-cognitive influences	Covers recent research on human factors	Emphasizes culture as weakest link in security	Limited technological innovation discussion
[71]	Proposes information security culture framework	Links culture to policy and technology integration	Reviews cultural factors over time	Provides comprehensive culture cultivation model	Limited direct technological innovation focus
[72]	Develops cybersecurity maturity assessment framework	Includes policy compliance and governance	Tracks maturity evolution in UK HEIs	Emphasizes organizational readiness and culture	Integrates multiple regulations and standards
[73]	Assesses risk management practices in Tanzanian HEIs	Highlights training and management gaps	Reviews recent risk management adoption	Notes cultural and management support importance	Limited technological innovation focus
[74]	Applies ISO 31,000 and ISSAF for risk assessment	Suggests training to improve risk mitigation	Reviews recent security incidents and responses	Emphasizes organizational policy and culture	Provides technical risk assessment recommendations
[75]	Discusses cybersecurity leadership and policy	Highlights training and compliance importance	Reviews leadership evolution in HEIs	Focuses on organizational psychology and culture	Includes AI and digital forensics tools
[76]	Applies situational crime prevention to cybersecurity	Links policy measures to crime reduction	Reviews recent crime prevention strategies	Emphasizes organizational controls and culture	Discusses firewall and access control technologies
[33]	Reviews policy compliance and culture	Highlights awareness and training roles	Covers policy compliance trends	Emphasizes organizational culture impact	Limited technological innovation focus
[77]	Investigates staff awareness and practices	Links training and policy to behavior	Reviews recent awareness levels	Highlights organizational culture and behavior	Limited technological innovation focus
[78]	Compares cybersecurity awareness in two countries	Measures training impact on behavior	Reviews recent pandemic-related changes	Notes cultural differences in behavior	Limited technological innovation focus

TABLE 1: Continued.

Study	Policy frameworks	Training effectiveness	Historical evolution	Organizational culture influence	Technological innovations
[79]	Discusses cybersecurity role in education	Emphasizes policy and training development	Reviews recent strategy development	Highlights culture of digital hygiene	Limited technological innovation focus
[24]	Case study on policy compliance factors	Links justice and culture to compliance	Reviews policy adherence evolution	Emphasizes organizational culture and sanctions	Limited technological innovation focus
[80]	Describes human-centric cybersecurity program	Focuses on training and management	Reviews program development and outcomes	Addresses organizational culture	Limited technological innovation focus
[81]	Proposes cybersecurity framework for university systems	Emphasizes policy and governance	Reviews cloud system security evolution	Highlights organizational culture and training	Integrates encryption, monitoring, AI, blockchain
[22]	Examines human error in data breaches	Links training and policy to error reduction	Reviews recent breach incidents	Emphasizes culture and awareness	Limited technological innovation focus
[82]	Proposes lightweight cyber risk framework	Highlights policy gaps and proactive controls	Reviews recent risk management trends	Notes organizational and cultural challenges	Uses penetration testing and proactive controls
[83]	Presents data security mechanisms for online platforms	Emphasizes policy and training for protection	Reviews recent cybercrime incidents	Highlights organizational culture and awareness	Suggests security mechanism adoption
[42]	Systematic review of IS management frameworks	Identifies standards and strategies	Reviews ISM framework evolution	Emphasizes governance and culture	Covers ISO, COBIT, ITIL, NIST standards
[84]	Assesses cybersecurity posture in Tanzanian HEIs	Notes policy and training deficiencies	Reviews recent preparedness levels	Highlights organizational culture and vigilance	Limited technological innovation focus

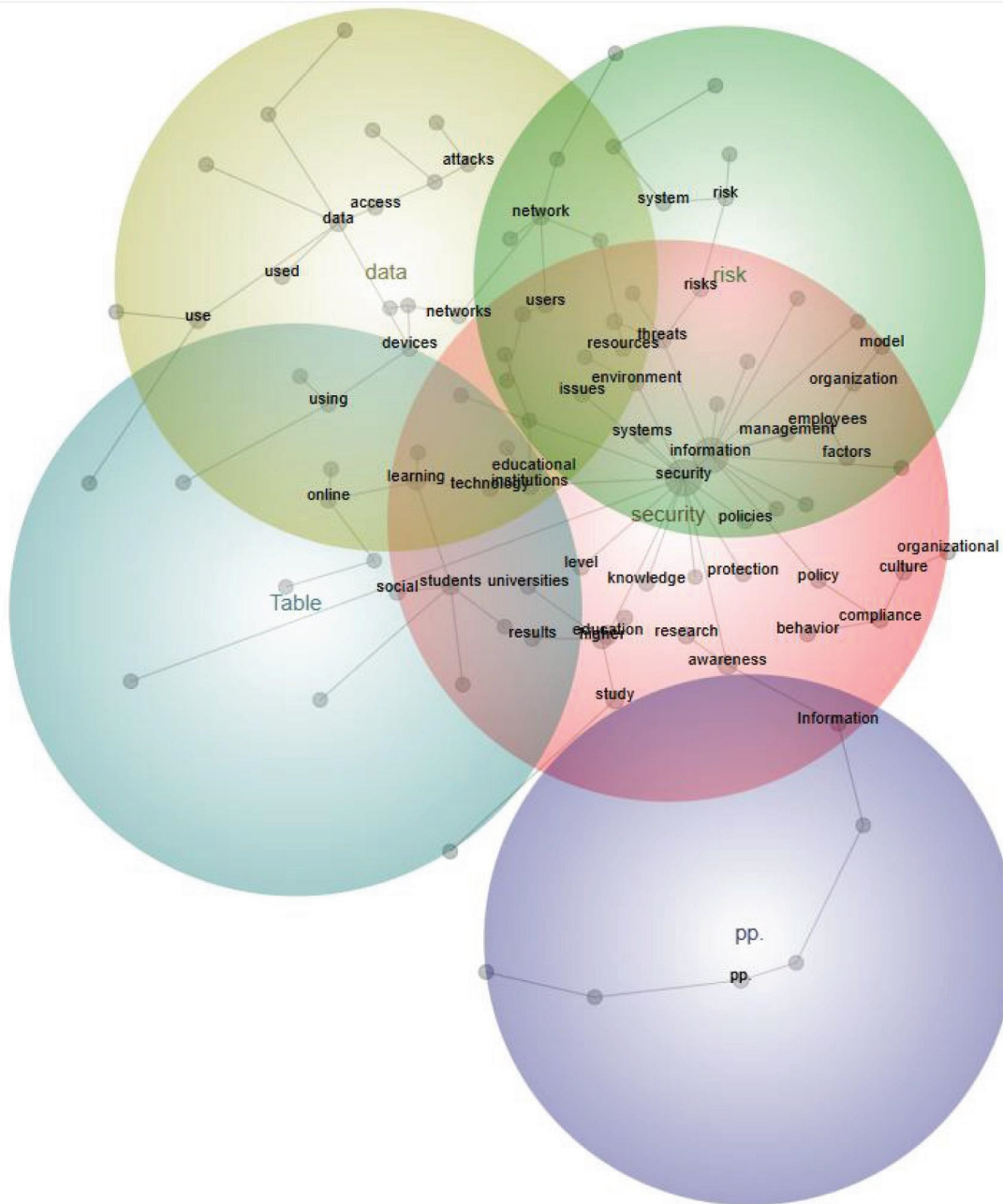


FIGURE 2: Concept map created using Leximancer. Source: author's creation.

Several studies also underscore the importance of integrating training with robust policy enforcement and advanced technological safeguards to maximize its impact [51, 63]. However, a common challenge is that awareness programs frequently contend with inadequate resources or inconsistent implementation, especially in institutions with limited funding, thereby diminishing their overall efficacy [4, 61]. Also, despite security policies being in place, many HEIs lack effective awareness and training programs specifically adapted to their unique needs [8]. Moreover, the absence of standardized metrics for evaluating training effectiveness often hinders the accurate assessment of return on investment and

identification of areas for improvement [19]. This gap often leads to employees having low policy awareness, which can result in noncompliant behavior, rendering even robust policies ineffective [89]. Therefore, consistent commitment from institutional leadership to foster a security-conscious culture through ongoing education can transform employees from potential vulnerabilities into formidable assets against cyber threats [86, 90].

5.3. RQ.3 How Have Information Security Practices in HEIs Progressed Across Different Historical Phases? Twenty-eight distinct studies document the historical progression of

information security practices within HEIs, showcasing a shift from rudimentary technical controls to sophisticated, integrated management frameworks that encompass policy development, cultural integration, and technological safeguards [3, 20, 91]. This evolution has been significantly propelled by the transition to digital and cloud-based learning environments, necessitating the rapid adoption of advanced security measures and comprehensive awareness initiatives [49, 58, 62].

Furthermore, historical analyses consistently highlight persistent difficulties in harmonizing policy, training, and organizational culture, although gradual improvements in governance and risk management have been observed [53, 67]. More recently, the advent of the COVID-19 pandemic and the widespread adoption of hybrid learning models have introduced novel vulnerabilities, prompting security adaptations [1, 78]. Consequently, contemporary trends now incorporate the integration of artificial intelligence, blockchain technology, and behavioral analytics to effectively counter evolving threats [56, 74]. This ongoing transformation mandates a proactive and adaptable approach to information security within higher education, emphasizing continuous threat intelligence and dynamic policy adjustments to effectively combat sophisticated cyber adversaries [9].

Moreover, the increasing intricacy of regulatory compliance and data privacy concerns further obliges HEIs to adopt a comprehensive, multilayered security posture [18]. This strategy not only involves technological defenses but also cultivates a robust cybersecurity culture through continuous education and bespoke training programs for all stakeholders, thereby transcending mere technical solutions to recognize human factors as pivotal components of information security [8, 61, 92].

5.4. RQ.4 What Role Does Organizational Culture Play in Shaping Cybersecurity Behavior and Policy Compliance Within HEIs? Organizational culture significantly influences cybersecurity adherence, user behavior, policy compliance, and the effectiveness of training, as identified in 38 studies. Consequently, security culture readiness models and frameworks advocate for fostering internalized responsibility and collective vigilance among both staff and students [93]. Human error and insider threats are frequently linked to cultural weaknesses, underscoring the importance of awareness and behavioral interventions [1, 19, 22]. Thus, collaborative and human-centric approaches to cybersecurity management enhance cultural strength and resilience [50, 63].

However, cultural challenges often include resistance to policy enforcement, a lack of engagement, and varied perceptions of risk across institutional roles [24, 77]. This complexity necessitates a nuanced approach to cultivating a robust cybersecurity culture, one that transcends mere technical solutions to address the socio-technical dimensions of information security within higher education environments [93]. Indeed, this oversight can create vulnerabilities that technology alone cannot mitigate, as human behavior often represents the weakest link in the security chain [19].

Therefore, there is a critical need for HEIs to implement comprehensive information security policies that specifically

account for human behavior and foster a strong security culture [86, 94]. Establishing an information security culture is vital for organizational survival and success, particularly because employees are frequently identified as a significant source of security breaches due to negligence or incompetence [85]. This approach ensures that while technological solutions are crucial, the human element encompassing employee attitudes, awareness, and compliance with security protocols is equally prioritized to safeguard information assets [95].

5.5. RQ.5 How Are Emerging Technological Innovations Transforming Cybersecurity Practices in HEIs? Recent scholarship, encompassing 25 studies, underscores significant technological advancements in cybersecurity pertinent to HEIs. These innovations specifically involve AI-driven threat detection, blockchain for ensuring data integrity, biometric authentication, and sophisticated encryption methods [12, 56, 81]. Moreover, contemporary tools facilitate incident management, continuous monitoring, and behavioral analytics, all meticulously designed to bolster security postures [54, 74].

Furthermore, gamification and simulation technologies are increasingly deployed to enhance the efficacy and engagement of training programs [25, 69]. Despite this notable technological progression, many institutions grapple with adoption hurdles stemming from interoperability issues, budgetary constraints, and user acceptance challenges [56, 82].

Additionally, novel methodologies such as adaptive security systems and behavioral biometrics are under exploration in HEIs to dynamically adjust protective measures based on real-time risk assessments and user behaviors, thereby augmenting security without compromising usability [96]. Such advancements mandate a continuous re-evaluation of existing security frameworks to integrate cutting-edge solutions while concurrently addressing the distinct challenges inherent in academic environments [91]. For example, the integration of artificial intelligence and machine learning algorithms is profoundly transforming threat detection by enabling predictive analysis and automated response mechanisms, thus moving beyond conventional signature-based detection to identify emergent attack patterns [97].

Therefore, the escalating sophistication of cyber threats, especially those leveraging AI, further necessitates the development of robust, adaptive security systems capable of identifying and neutralizing complex, constantly evolving attack vectors [1, 98]. Notably, AI-enhanced solutions offer substantial improvements, including real-time threat detection, scalability, and adaptive security, rendering them critical for safeguarding sensitive financial and research data against emerging threats [99]. Furthermore, advanced intelligent systems and specialized software are continuously being developed and updated to enhance organizational data protection and efficiency across various sectors, including higher education, by reducing costs and strengthening virtual security measures [98].

6. Thematic Review and Comparative Analysis

6.1. Thematic Review. The existing body of literature regarding information security practices in HEIs reveals several

TABLE 2: Thematic review of information security practices in HEIs.

Theme	Subthemes	Key insights
Cybersecurity policies and frameworks	1. Development of robust policies 2. Implementation and enforcement 3. Integration with risk management	Research consistently emphasizes the indispensable role of comprehensive cybersecurity policies and frameworks, meticulously crafted to align with the distinct operational environment of HEIs. This study highlights the significance of frameworks that integrate robust governance structures, effective risk management protocols, stringent technical controls, and agile incident response mechanisms to cultivate formidable security postures. The incorporation of recognized standards, such as ISO 27000 and NIST, is frequently observed, establishing a foundational element for strategic cybersecurity planning and subsequent implementation. These frameworks are inherently designed for continuous evolution, enabling them to adapt proactively to emergent threats, escalating system complexities, and the intricate governance challenges characteristic of academic institutions
	4. Alignment with compliance standards	
User awareness and training	1. Security-conscious culture building 2. Ongoing training initiatives 3. Addressing human-factor vulnerabilities	User awareness and security training programs are identified as pivotal in reducing human vulnerability to cyber threats. Findings from this study report varied training methodologies, including gamification, simulated phishing, and targeted malware awareness programs, showing significant improvements in cybersecurity behaviors and knowledge among staff and students. Challenges remain in sustaining awareness over time and tailoring training to diverse user demographics and roles. Enhanced training is linked to improved policy compliance and reduced risk incidents
Data protection and technical controls	1. Technical controls (MFA, encryption, access management, and monitoring) 2. Governance structures (policy oversight and accountability) 3. Technological countermeasures	This study shows that data protection within HEIs is addressed through encryption, access controls, intrusion detection, and risk management strategies. Accordingly, Papers discuss the application of advanced technologies, such as AI, blockchain, and biometric authentication, to secure sensitive educational data and cloud-based systems. However, despite technological advances, vulnerabilities persist due to shadow IT, mobile device usage, and system decentralization, underscoring the need for continuous monitoring and adaptive security measures
Organizational culture and human factors	1. Influence of culture on policy adherence 2. Behavioral aspects of cybersecurity practices 3. Need for holistic, adaptable approaches	This study reveals that behavioral intentions, awareness, internalized responsibility, and leadership engagement impact compliance and resilience. Studies advocate for cultivating a security culture readiness model and addressing socio-cognitive factors to strengthen organizational defenses. However, human error remains a leading cause of breaches, necessitating focused interventions on education and culture transformation

interconnected and critical themes. Foremost among these is the foundational role of developing and implementing robust cybersecurity policies and frameworks to ensure effective risk management and data protection. Concurrently, user awareness and comprehensive training initiatives are consistently emphasized as pivotal for fostering a security-conscious culture and mitigating human-factor vulnerabilities. Data protection strategies, which encompass both technical controls and governance structures, are analyzed in the context of evolving threats and the application of innovative technological countermeasures. Ultimately, organizational culture and various human factors are identified as significantly influencing policy adherence and overall cybersecurity behavior, thereby highlighting the imperative for holistic and adaptable security approaches within academic settings. Table 2 outlines the identified themes of information security practices in HEIs.

6.2. Comparative Analysis of Information Security Practices in HEIs. The reviewed literature generally agrees on the critical importance of robust information security policies and the need for continuous user awareness training tailored to HEIs. Many studies highlight organizational culture and human factors as pivotal in shaping effective cybersecurity behavior, emphasizing the necessity of fostering a security-aware environment. Divergences arise, however, particularly regarding the effectiveness of specific policy measures, technological innovations, and the extent of policy implementation across different regional and institutional contexts. These discrepancies often stem from variations in research designs, geographic focus, institutional maturity, and the evolving technological landscape, as discussed in this study.

Multiple studies emphasize the critical role of comprehensive and adaptable information security policies within HEIs. They highlight governance, risk management, and clear

guidelines as essential for effective cybersecurity [12, 53, 60, 67, 87]. There is also a consensus that policies must be flexible to accommodate institutional culture and technological changes. However, some studies note gaps or weaknesses in policy implementation and coverage, specifically mentioning limited policy presence or enforcement challenges in certain regions or institutions [18, 62, 82]. Furthermore, some papers report that policies alone do not guarantee compliance or awareness [57]. This discrepancy may result from differences in institutional maturity, regional resources, and the dynamic nature of HEI environments. Variations in policy enforcement mechanisms and institutional commitment levels also contribute to differing outcomes.

Second, there is a widespread consensus that user awareness training substantially enhances cybersecurity compliance and mitigates risky behaviors among faculty and students [21, 25, 51, 64, 68, 77]. Specifically, interactive and targeted initiatives, such as simulated phishing exercises, prove particularly efficacious [25, 64]. Such training is instrumental in cultivating a robust security culture and promoting desirable behavioral modifications [19, 55]. Nevertheless, several studies identify a discrepancy between awareness and actual behavior, suggesting that knowledge does not consistently translate into adherence [24, 57, 66]. Moreover, the efficacy of training can fluctuate depending on delivery modalities, cultural determinants, and engagement metrics [78]. Persistent challenges include sustaining consistent training curricula and precisely quantifying their enduring impact [18, 22]. These inconsistencies often stem from variations in study demographics, cultural contexts, and the evaluative methodologies employed for training. Furthermore, differing degrees of institutional endorsement and the innovative nature of programs significantly influence training effectiveness.

Several studies delineate the evolutionary trajectory of information security practices within HEIs, observing a notable transition from purely technology-centric defenses to comprehensive frameworks that incorporate human elements and organizational culture [3, 12, 20, 49]. The escalating complexity of cyber threats continually compels HEIs to cultivate more advanced security postures. Nevertheless, some research indicates a persistent delay in implementing exhaustive policies or adopting cutting-edge technologies, particularly in environments with limited resources [73, 82]. Other investigations underscore enduring challenges, such as vulnerabilities perpetuated by human error, despite historical advancements [22]. These disparities may arise from variations in regional development, financial allocations, and institutional strategic imperatives, all of which shape the pace and scope of historical progression. Divergences in research timelines and the rates of technological integration further contribute to this observed inconsistency.

Numerous studies consistently acknowledge the critical importance of organizational culture and human behavior in achieving effective cybersecurity. Research underscores that awareness, accountability, and ingrained security values are fundamental to ensuring adherence to policies and mitigating risks [18–20, 55, 70, 71]. Nevertheless, discrepancies

emerge concerning the extent to which cultural elements are integrated into security frameworks and their quantifiable impact on behavior. Some findings indicate inadequate incorporation of cultural aspects into security policies or training programs [24, 57, 62], and the influence of demographic variables exhibits inconsistency across different studies [21, 23, 52]. This variation can be attributed to differences in institutional maturity, cultural diversity, and the methodological approaches employed to investigate human factors. Furthermore, the level of institutional dedication to cultural transformation and various contextual factors also differ significantly.

Many studies emphasize the incorporation of nascent technologies, including artificial intelligence, blockchain, advanced encryption, and multifactor authentication, as pivotal for bolstering cybersecurity resilience within HEIs [4, 12, 56, 58, 74, 81]. The amalgamation of technological solutions with robust governance frameworks and comprehensive training programs is frequently regarded as a leading practice. Nevertheless, certain research identifies substantial implementation hurdles, such as issues of interoperability, privacy concerns, and budgetary limitations [56]. Furthermore, significant deficiencies in incident response protocols and penetration testing readiness are also observed [74, 82]. The effectiveness of encryption, moreover, remains a contentious topic due to documented vulnerabilities [76]. These disparities frequently arise from variations in institutional resources, technological preparedness, and diverse geographical contexts. Additionally, the swift pace of technological advancement implies that certain studies possess greater currency than others, contributing to divergent findings regarding the efficacy of various innovations.

7. Theoretical and Practical Implications

7.1. Theoretical Implications. The synthesis in this study underscores the critical role of human factors, organizational culture, and behavioral intentions in shaping information security compliance and cybersecurity resilience in HEIs. This supports and extends existing socio-cognitive and behavioral theories by emphasizing the interplay between institutional commitment, user awareness, and security behavior, highlighting gaps where awareness does not always translate into compliant behavior [18, 19, 57].

The reviewed frameworks and models, such as the security culture readiness model and human-centric cybersecurity frameworks, contribute to theoretical advancements by integrating organizational learning, diffusion of innovation, and technology-organization-environment perspectives. These frameworks provide a holistic understanding of cybersecurity readiness beyond technical controls, emphasizing culture and collaboration [50, 55, 59].

Findings related to cybersecurity policy development and compliance reinforce the importance of policy clarity, comprehensiveness, and flexibility, aligning with governance theories that stress procedural justice and organizational culture as determinants of policy adherence [24, 60].

The application of situational crime prevention theory to cybersecurity in HEIs offers a novel theoretical lens,

demonstrating that increasing effort, risk, and reducing rewards can effectively mitigate cybercrime. However, the limitations of certain technical controls, such as encryption vulnerabilities, challenge assumptions about their efficacy and call for theoretical refinement in cybersecurity prevention strategies [76].

The evolution of cybersecurity education, including gamification and malware awareness programs, supports educational theories that advocate for active, engaging learning methods to enhance knowledge retention and behavioral change in cybersecurity contexts [64, 69].

The integration of risk management standards (e.g., ISO 31000) with penetration testing frameworks in HEIs contributes to the theoretical discourse on combining technical and managerial approaches for comprehensive cybersecurity risk assessment [74, 87].

7.2. Practical Implications. HEIs must prioritize institutional commitment by establishing clear governance structures, continuous monitoring, and enforcement mechanisms to foster secure behaviors among staff and students. This includes regular audits, policy reviews, and addressing human factors that contribute to security breaches [24, 57, 60].

Cybersecurity awareness and training programs tailored to the diverse HEI populations are essential. Evidence suggests that targeted, engaging, and context-specific training, such as simulated phishing campaigns and malware awareness initiatives, significantly improves security knowledge and compliance [25, 51, 64].

Collaborative frameworks, such as EduCERT, demonstrate practical benefits in enhancing national and institutional cybersecurity resilience through knowledge sharing, incident management, and coordinated response efforts. Such models should be adapted and scaled globally to address diverse HEI environments [50].

The adoption of multilayered cybersecurity frameworks that integrate technical controls (e.g., firewalls, encryption, and multifactor authentication) with organizational policies and user education is critical. However, institutions must recognize the limitations of certain technical measures and continuously update defenses against evolving threats [4, 76].

Policymakers and HEI leaders should leverage comprehensive cybersecurity maturity and risk management frameworks to benchmark current practices, identify gaps, and implement strategic improvements aligned with international standards and best practices [72, 73, 82].

Emerging technologies, such as AI, blockchain, and biometric authentication, offer promising avenues for enhancing security in specialized HEI domains like libraries and cloud-based systems. Nonetheless, practical challenges, including interoperability, privacy concerns, and budget constraints, must be addressed for successful implementation [56, 58].

The findings advocate for a systemic approach that balances technical, human, and organizational dimensions of cybersecurity, emphasizing the need for continuous adaptation to new threats, fostering a culture of security awareness,

and ensuring compliance through effective leadership and policy enforcement [12, 19, 75].

8. Limitations of the Study

As presented in Table 3, the systematic literature review conducted for this study was subject to several limitations.

9. Conclusion, Research Gaps, and Future Directions

There are limited empirical studies that assess the real-world effectiveness of cybersecurity policies in HEIs, especially regarding their longitudinal impacts on compliance and security outcomes. Future studies should conduct longitudinal, multi-institutional research measuring policy implementation outcomes, compliance rates, and incident reduction over time, and develop metrics for policy effectiveness specifically tailored to HEIs. This is essential because existing frameworks, though comprehensive, often lack empirical validation, which limits our understanding of a policy's actual impact on security behavior and breach prevention [20, 60, 67].

This study also highlights significant gaps between cybersecurity knowledge gained through training and the actual secure behaviors among HEI users. It is recommended that future research develop and test adaptive, role-specific training programs incorporating behavioral science to close this knowledge-action gap. Further research should also evaluate long-term behavior change post-training. This recommendation stems from studies showing that awareness does not consistently translate into secure behavior, indicating a need for more effective, behaviorally informed training approaches [21, 23, 66].

Furthermore, while AI, blockchain, and biometric technologies are recognized as promising, their practical adoption and effectiveness in HEIs remain underexplored. Future studies should investigate barriers to adoption, interoperability challenges, and user acceptance of emerging cybersecurity technologies in HEIs. Pilot implementations with rigorous evaluation of both security and usability are highly recommended. This is because, despite the noted theoretical benefits of emerging cybersecurity technologies, their practical deployment is often limited by technical, financial, and cultural factors [12, 56, 81].

Also, this study observes that while organizational culture's influence on cybersecurity is acknowledged, quantitative, large-scale studies linking culture to security outcomes are scarce. There is a need to develop validated instruments to quantitatively measure security culture in HEIs and correlate findings with incident rates and compliance data. Research should also explore the effects of cultural diversity within institutions. This is crucial because current research on information security in HEIs relies heavily on qualitative or self-reported data, which limits generalizability and actionable insights [18–20].

This study reveals that many HEIs lack formal, integrated risk management frameworks capable of adapting to evolving cyber threats and institutional contexts. Therefore, it

TABLE 3: Limitations of the study.

Area of limitation	Description of limitation	Papers that have limitations
Small sample sizes	Several studies suffer from small or unrepresentative sample sizes, limiting the external validity and generalizability of their findings. This methodological constraint reduces confidence in applying results broadly across diverse higher education contexts	[57, 76]
Geographic bias	Research is often concentrated in specific regions or countries, which restricts the applicability of findings globally. This geographic bias limits understanding of diverse institutional and cultural cybersecurity challenges in higher education worldwide	[21, 55, 58]
Cross-sectional design	Many studies utilize cross-sectional data, which impedes the ability to establish causal relationships or temporal ordering between cybersecurity practices and outcomes. This limits the depth of insights into the evolution and effectiveness of interventions over time	[57, 76]
Limited focus on human factors	Despite recognition of human factors as critical in cybersecurity, there remains insufficient empirical exploration of behavioral and cultural influences on security compliance and awareness, weakening comprehensive understanding of organizational cybersecurity dynamics	[18–20, 59]
Low response rates	Surveys in some studies report low response rates, raising concerns about response bias and the representativeness of the data. This limitation undermines the reliability of conclusions drawn about cybersecurity awareness and behavior in higher education populations	[57]
Lack of longitudinal studies	The scarcity of longitudinal research restricts understanding of how cybersecurity policies, awareness, and behaviors develop and change over time, limiting the ability to assess long-term effectiveness of interventions in higher education institutions	[76]
Overemphasis on technical solutions	A predominant focus on technological measures overshadows the importance of organizational, cultural, and human-centric approaches, resulting in incomplete frameworks that may neglect critical socio-technical aspects necessary for effective cybersecurity management in academia	[12, 62, 70]
Limited diversity in institutional types	Most research centers on universities, with less attention to other higher education institutions, such as colleges or vocational schools, which may face distinct cybersecurity challenges, thus limiting the comprehensiveness of the literature	[60, 82]
Insufficient evaluation of training effectiveness	While many studies advocate for user awareness training, few rigorously evaluate the effectiveness of different training methodologies, leaving gaps in evidence-based best practices for enhancing cybersecurity compliance and resilience	[25, 64, 68]
Data age and relevance	Some studies rely on outdated data or pre-pandemic contexts, which may not accurately reflect the current cybersecurity threat landscape or the impact of recent technological and organizational changes in higher education environments	[66, 76]

recommends developing and validating lightweight, scalable risk management frameworks specifically tailored for HEIs, which should incorporate continuous risk assessment and incident response planning. The current inconsistency in risk management practices and their frequent detachment from institutional strategies contribute significantly to increased vulnerability [73, 74, 82].

In line with the foregoing discussion, shadow IT introduces significant security risks in HEIs; however, research on its effective management and policy integration remains limited. This study recommends conducting empirical studies on shadow IT prevalence, associated risks, and user motivations within HEIs. Furthermore, HEIs should develop robust governance models and comprehensive awareness programs to manage shadow IT responsibly. This approach is crucial because while shadow IT is prevalent due to flexible academic needs, it simultaneously creates visibility and control gaps, thereby increasing potential attack surfaces and information security vulnerabilities in HEIs.

It has also emerged that training programs on information security often lack customization for demographic, cultural, and role-based differences among HEIs stakeholders. This

study suggests research on the impact of tailored training content and delivery methods on diverse user groups, including students, faculty, and administrative staff, across cultures. This suggestion is justified by the fact that demographic and cultural factors, though underexplored, significantly influence training effectiveness and security behavior [21, 23, 52].

Collaborative frameworks, such as EduCERT, exhibit considerable potential; however, their extensibility, flexibility, and overall influence across disparate international higher education landscapes warrant a comprehensive assessment. This necessitates the execution of comparative case studies and pilot initiatives that deploy collaborative frameworks within diverse geographical and institutional environments, evaluating both their successes and inherent difficulties. These practical implementation challenges and contextual variances currently constrain the widespread acceptance and efficacy of these collaborative models [50].

The emerging use of behavioral analytics to detect complex cyber threats in HEIs is promising, yet it currently lacks extensive validation and integration into operational security. Therefore, it is recommended that future research develops and tests behavioral analytics tools using digital footprint and

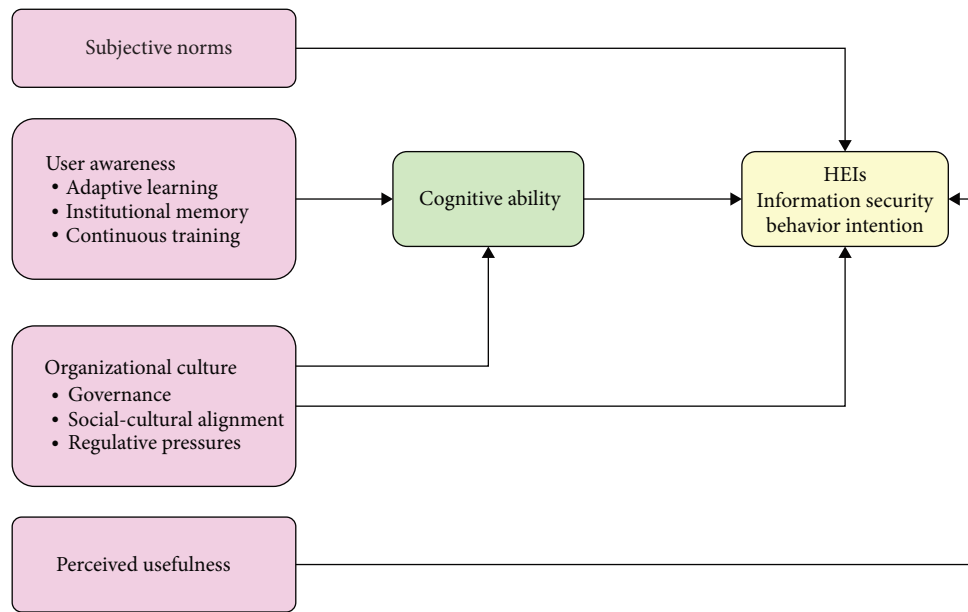


FIGURE 3: Information security conceptual framework.

sequential pattern analysis to identify insider threats and anomalous behaviors within HEIs. While early studies suggest improved threat detection capabilities, their operational integration and overall efficacy still require further demonstration [54].

The long-term impact of COVID-19 and hybrid learning on cybersecurity practices and vulnerabilities in HEIs is not yet fully understood. Therefore, longitudinal research is needed to track changes in cybersecurity policies, user behavior, and incident rates in HEIs before, during, and after the pandemic. This understanding is critical for developing adaptive security strategies, especially given that rapid shifts to remote learning introduced new vulnerabilities [1, 62, 78].

9.1. Conceptual Framework. This study advances a conceptual model illustrated in Figure 3. It integrates three theoretical lenses: institutional theory, organizational learning theory, and the technology acceptance model (TAM) to provide a holistic framework for advancing information security in HEIs. The model emphasizes the interplay between organizational context, learning processes, socio-technical alignment, and user acceptance of security technologies in building institutional resilience. Institutional pressures and governance structures establish the foundation for policy and cultural alignment. Organizational learning mechanisms enable adaptive improvement through continuous training and feedback. Finally, technology acceptance dynamics influenced by perceived usefulness determine the adoption and sustained utilization of security innovations. Together, these interdependent elements foster institutional resilience and long-term adaptability in managing cybersecurity threats.

9.2. Importance of the Conceptual Model. The conceptual model is important because it provides an integrative framework for understanding and improving information security practices in HEIs. It bridges multiple theoretical perspectives,

socio-technical systems (STS) theory, institutional theory, organizational learning theory, and the TAM to capture the multidimensional nature of cybersecurity. By combining these lenses, the model moves beyond narrow technical or compliance-oriented approaches and emphasizes a balanced, systemic view of security.

First, the model underscores that effective information security in HEIs depends on alignment between people, processes, and technology. This socio-technical balance ensures that security measures are not only technically sound but also culturally and operationally feasible within academic environments that value openness and collaboration.

Second, the model highlights the role of institutional structures and governance in shaping behavior. By integrating institutional theory, it shows how policies, norms, and cultural expectations create both the formal and informal boundaries within which security practices evolve.

Third, the inclusion of organizational learning theory emphasizes that resilience in cybersecurity is a continuous process, built through feedback, adaptation, and shared learning across departments and user groups. This fosters institutional memory and responsiveness to emerging threats.

Fourth, incorporating the TAM draws attention to the human factors influencing technology adoption. It explains why even sophisticated tools, such as AI-driven detection systems or blockchain-based records, can fail without user trust, perceived usefulness, and ease of use.

Overall, the model's importance lies in its capacity to guide both researchers and practitioners toward holistic, adaptive, and human-centered cybersecurity strategies. It enables HEIs to design interventions that harmonize governance, training, culture, and technology, thus strengthening compliance, reducing vulnerabilities, and promoting a culture of shared responsibility for information security. The model, therefore, serves not only as a theoretical contribution but also as a

practical roadmap for achieving sustainable cybersecurity resilience in academic institutions.

The author employed the Leximancer tool to represent article themes visually and utilized Silvi.ai for article categorization, including rationale for acceptance or rejection. Based on pre-established exclusion and inclusion criteria, two independent reviewers assessed articles, first by title and abstract, then by full text. Key findings were subsequently extracted and organized in MS Excel for inclusion in the study. The author reviewed and refined the content as necessary, assuming full responsibility for the publication. This rigorous methodology ensures a comprehensive and unbiased analysis of information security vulnerabilities within higher education, providing a solid foundation for subsequent discussion and recommendations.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Conflicts of Interest

The authors declare no conflicts of interest.

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